

(TR-104) PYTHON WITH AI/ML

Training Day 55 Report

Date: 27 March 2026

Topic Covered: Google Cloud Workflows & Workflow Orchestration Concepts

The fifty-fifth day of training focused on understanding Google Cloud Workflows, workflow orchestration concepts, execution control mechanisms, and different workflow implementation approaches used in modern automation systems.

- Explored Google Cloud Workflows as a fully managed orchestration service for automating multi-step processes.
- Learned how workflows execute services in a predefined sequence as part of an automated pipeline.
- Explored integration of workflows with Cloud Run, Cloud Functions, Google Cloud services, and external HTTP APIs.
- Understood workflow architecture including workflow definition, deployment, and execution phases.
- Learned how workflows function as serverless orchestration systems for managing distributed tasks.
- Studied important workflow characteristics such as scalability, reliability, and cost-efficient execution without idle resource usage.
- Learned sequential execution of workflow steps for ordered task processing.
- Explored conditional logic for implementing decision-based execution flows.
- Understood iteration concepts for processing lists, collections, and repeated operations inside workflows.
- Learned parallel execution techniques for running multiple workflow steps concurrently.
- Explored subworkflows for creating modular and reusable workflow components.
- Studied different categories of workflow systems including code-based workflow tools and visual workflow tools.

- Explored platforms that support both code-based and visual workflow development approaches.
- Strengthened understanding of workflow lifecycle management and orchestration strategies used in cloud-based automation systems.
- Improved understanding of scalable workflow design concepts for AI and automation applications.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/agents>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/google-cloud-workflows-automating-cloud-services-with-smart-orchestration-21ef87ba30d4>